ESTHER WANJIKU MAINA

GEOSPATIAL & REMOTE SENSING EXPERT

CONTACT

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ABOUT ME

Esther is a remote sensing expert specialising in food systems and natural resource management. She leads projects that focus on developing innovative solutions for geospatial big data analytics and visualisation, supporting data-driven decision-making. She has a background in Geospatial Information Science, and also certified AWS Cloud Practitioner and is an active member of the International Society for Photogrammetry and Remote Sensing (ISPRS), African Women in GIS, and a graduate member of the Institution of Surveyors of Kenya. Her work bridges technology and development, with a strong focus on applying Earth Observation and AI tools for sustainable impact.

SKILL

- GIS
- Remote Sensing
- Data management
- Web development
- Project Management

LANGUANGE

English

Swahili

AWARD

IAF Launchpad Mentee 2023
IAF

Space4Women Mentee 2023
UNOOSA

UN/Austria Symposium presentation acceptance and Travel Grant 2023
UNOOSA

UNOOSA Space4Water presentation acceptance and Travel Grant 2025
UNOOSA

EXPERIENCE

2022-To date **REMOTE SENSING OFFICER**
Kenya Space Agency

- Space Innovation Roadmap for the Space sector in Kenya: May 2024-To date, Technical Analyst
- Agricultural Project: July 2023-To date, Lead Remote Sensing KSA team in the project
- Flood Risk Analysis: March 2022-Dec 2023, technical lead in the project
- Education and Outreach: Nov 2023-To date, Earth Observation Thematic Lead
- Developing data quality protocols and standards for internal data processes
- Support in the development of policies and strategies related to Research, Science, Technology, and Innovation and Earth Observation

OCt 2024-Jan 2025 **CONSULTANT - COMMUNITY MANAGER**
DevGlobal

- Engaged and expanded a diverse ICT4Ag community through targeted outreach.
- Managed digital platforms to enhance user interaction and knowledge exchange.
- Developed and shared content to promote learning in digital agriculture.
- Planned and facilitated webinars and virtual events with regional participation.
- Supported advocacy efforts by coordinating policy dialogues and expert discussions.

2016 - 2018 **AZIMATH LIMITED**
GIS Intern

- Validated the field data and analysed field data under the KPLC Meter Box campaign;
- Digitized cadastral sheets and aerial images; and
- Together with the cadastral team, we developed spatial plans for Uasin Gishu and Kisumu counties.

EDUCATION

2024 – 2025 European Institute of Innovation for Sustainability, Italy
Master's in Space Entrepreneurship

2012- 2016 Dedan Kimathi University of Technology, Kenya
Bachelor of Science, Geospatial Information Science

2008-2011 Ng'araria Girls Secondary School
Kenya Certificate of Secondary Education; attained a mean grade of B, 63 points

1999-2007 Kaimiri Primary School, Kenya Certificate of Primary School

COURSES AND TRAINING

2025 STI mainstreaming in MDAs in rapidly changing times - NACOSTI
Project Management – European Business Institute Luxembourg
2024 Managing Successful Field Research - World Bank Development Impact
Online Summer School on Astronomy – FAST4Future
Humanitarian Simulation – NASA Lifeline
Agriculture Informatics using GIS and RS – IIRS, ISRO
AWS Cloud Practitioner Certified – Amazon Web Services
School of Climate Change – Oxford Climate Society
Basic remote sensing for educators – Junior Academy of Sciences of Ukraine
Report Writing – Kenya School of Government
Supervisory skills – Management University of Africa
2023 Reproducible Research Fundamentals – World Bank Development Impact
Space Law for New Space Actors – United Nations E-Learning
AI & Space Summer Course - Amsterdam Law and Technology Institute
Women and Leadership - European Business School Luxembourg
Software Engineering - Moringa School
2022 Mapping crops - ARSET
Introduction to Sandbox - Digital Earth Africa
2021 Python for Everyone - University of Michigan -Coursera
2020 SQL Fundamentals - Data Camp
R Programming - Udemy
2017 Certificate in Basic Kenya Sign Language - ACK All Saints, Nairobi
2012 Computer packages - ACK All Souls College, Machakos

A registered member of:

- Institution of Surveyors of Kenya - Graduate Member, Geospatial Chapter
- International Society for Photogrammetry and Remote Sensing (ISPRS)
- African women in GIS
- Coding black females
- GMES & Africa

VOLUNTEER

July 2023 – Dec 2024 *Technical Assistant* *African Space Leadership Institute*
Voluntereed with the ASLI team as a technical assistant volunteer, my responsibilities include:

- Creating and populating a space policy inventory;
- Technical support in seminars, dialogues, and conferences; and
- Support in developing training schedules and trainings coordination.

Feb – May 2024 *ICT4Ag Ambassador* *DevGlobal*

- Supported the ICT4Ag Nairobi event by mobilizing participants, leading social media advocacy, facilitating discussions, and prepared the event repor.

July 2023 – October 2023 *Youth Jury Volunteer* *Danone Ramsar Convention*

- Participated in the Youth Jury for the Danone-Ramsar Action for Wetlands Fund, reviewing grant proposals aimed at supporting innovative solutions to conserve, restore, and sustainably manage wetlands.

May-Aug. 2012 *Voluntary Teaching* *Kiugu Secondary School*

- Taught English, mathematics, and physics subjects.

REFEREEES

Available upon request